413/613 Fall 2026 - Applied Statistics & Data Science I - Lecture 14

Prof. Suryanarayana Maddu
Department of Applied Mathematics and Statistics
Data Science and AI Institute
Johns Hopkins University

Recall

In Lectures 12 and 13, we considered linear regression models of the form

$$\mathbf{y} = X\beta + \varepsilon,$$

with ordinary least-squares estimator

$$\boxed{\hat{\beta} = (X^T X)^{-1} X^T \mathbf{y}.}$$

We also examined how to assess a fitted model using residuals, R^2, leverage, collinearity, and the conditioning of $X^T X$.

A new question now arises:

$$\boxed{\text{How complex should the regression model be?}}$$

A model that is too simple may fail to capture systematic structure in the data, while a model that is too flexible may fit the observed sample well but behave poorly on new data.

1 Nonlinear Models from Linear Regression

In simple linear regression, we assumed that the conditional mean of the response is linear in the predictor:

$$\mu(x) = \mathbb{E}[Y \mid X = x] = \beta_0 + \beta_1 x.$$

This model can only represent a straight-line (linear) relationship between X and the expected response.

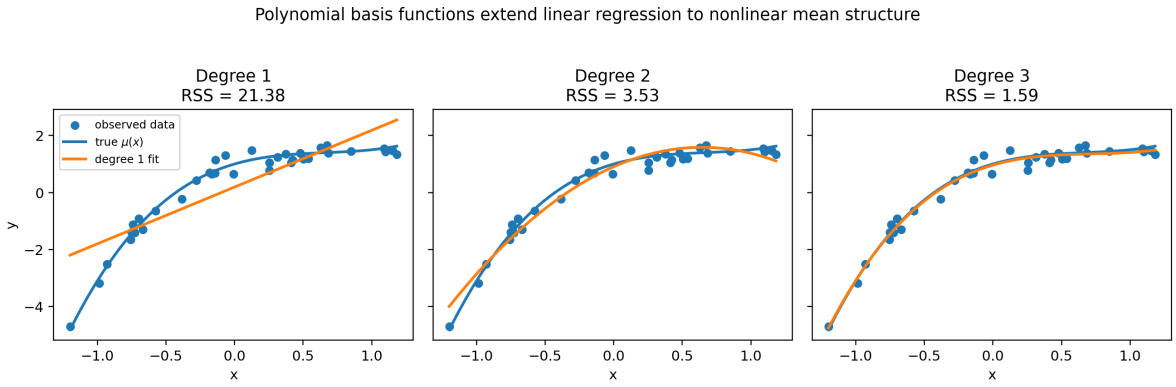


Figure 1: Polynomial basis expansion applied to the same nonlinear dataset. A degree-1 model is too restrictive to represent the curvature in the conditional mean. Adding the basis function x^2 substantially improves the fit, while adding x^3 allows the model to capture the remaining nonlinear structure. In every case, the model remains linear in the coefficients β_j.

Suppose instead that the observed data suggest a curved (non-linear) relationship as shown in Fig. 1. Then a linear conditional mean may be too restrictive, as illustrated in the left panel of Fig. 1. One way to increase

the flexibility of the model is to introduce nonlinear functions of the predictor. For example, a quadratic model is

$$\mu(x) = \beta_0 + \beta_1 x + \beta_2 x^2.$$

More generally, a polynomial model of degree p is

$$\mu(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_p x^p.$$

The corresponding regression model is

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \cdots + \beta_p X^p + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid X] = 0.$$

This model is nonlinear in the predictor X, but it is still linear in the unknown coefficients

$$\beta_0, \beta_1, \ldots, \beta_p.$$

This distinction is important: ordinary least squares requires the model to be linear in the unknown parameters, not necessarily linear in the predictor.

We can make this explicit by defining new features

$$\phi_0(x) = 1, \qquad \phi_1(x) = x, \qquad \phi_2(x) = x^2, \qquad \ldots, \qquad \phi_p(x) = x^p.$$

Then

$$\mu(x) = \sum_{j=0}^{p} \beta_j \phi_j(x).$$

The functions

$$\phi_0, \phi_1, \ldots, \phi_p$$

are called *basis functions*. Polynomial regression therefore amounts to replacing the original predictor x by the feature vector

$$\phi(x) = \begin{pmatrix} 1 \\ x \\ x^2 \\ \vdots \\ x^p \end{pmatrix} \in \mathbb{R}^{(p+1)\times 1}.$$

For observations

$$(x_1, y_1), \ldots, (x_n, y_n),$$

the design matrix becomes

$$X = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^p \\ 1 & x_2 & x_2^2 & \cdots & x_2^p \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^p \end{pmatrix} \in \mathbb{R}^{n\times(p+1)}.$$

The coefficient vector is

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix} \in \mathbb{R}^{(p+1)\times 1},$$

and the response vector is

$$\mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^{n \times 1}.$$

Therefore, despite the nonlinear dependence on x, the model retains exactly the same matrix form as before:

$$\mathbf{y} = X\beta + \varepsilon.$$

Consequently, ordinary least squares still gives

$$\hat{\beta} = (X^T X)^{-1} X^T \mathbf{y},$$

provided $X^T X$ is invertible.
Increasing the polynomial degree therefore increases the number of basis functions and enlarges the class of conditional mean functions that can be represented. This increased flexibility motivates the next question:

How complex should the model be?

2 Model Complexity

In the previous section, we modeled the conditional mean using a polynomial basis expansion,

$$\mu_p(x) = \mathbb{E}[Y \mid X = x] = \beta_0 + \beta_1 x + \cdots + \beta_p x^p.$$

Increasing the polynomial degree p increases the number of basis functions

$$1,\ x,\ x^2,\ \ldots,\ x^p,$$

and therefore increases the flexibility of the model. For example,

$$\mu_1(x) = \beta_0 + \beta_1 x,$$

$$\mu_2(x) = \beta_0 + \beta_1 x + \beta_2 x^2,$$

while

$$\mu_{14}(x) = \beta_0 + \beta_1 x + \cdots + \beta_{14} x^{14}$$

can represent a much larger class of nonlinear functions.

Let

$$\hat{\mu}_p(x)$$

denote the degree-p polynomial fitted by ordinary least squares. For the observed data

$$(x_1, y_1), \ldots, (x_n, y_n),$$

the fitted values are

$$\hat{y}_i = \hat{\mu}_p(x_i),$$

and the training residual sum of squares is

$$\boxed{RSS_p = \sum_{i=1}^{n} \left(y_i - \hat{\mu}_p(x_i)\right)^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2.}$$

The degree-p model is contained within the degree-$(p+1)$ model, because setting

$$\beta_{p+1} = 0$$

reduces

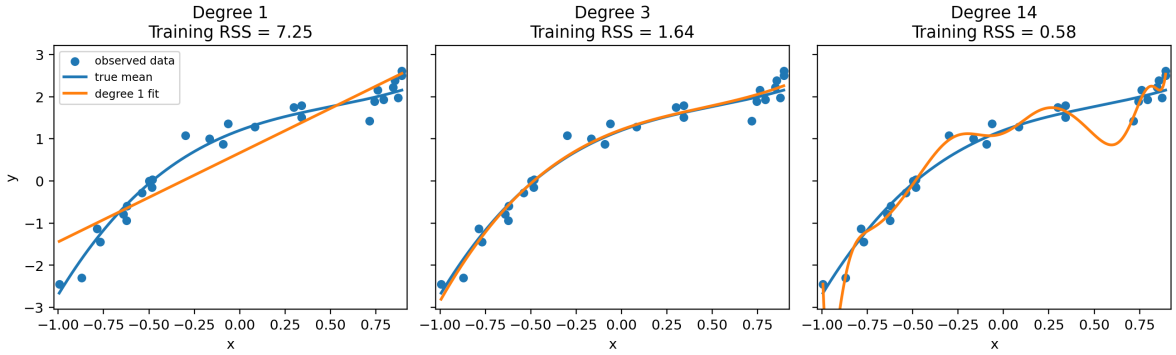


Figure 2: Effect of polynomial degree on model flexibility. The degree-1 model is too restrictive to capture the nonlinear structure in the conditional mean. The degree-3 model captures the underlying trend well, while the degree-14 model achieves a smaller training RSS by fitting additional sample-specific fluctuations.

$$\beta_0 + \beta_1 x + \cdots + \beta_p x^p + \beta_{p+1} x^{p+1}$$

to the degree-p model. Therefore, the degree-$(p+1)$ model minimizes the residual sum of squares over a larger class of candidate functions. Hence,

$$RSS_{p+1} \leq RSS_p.$$

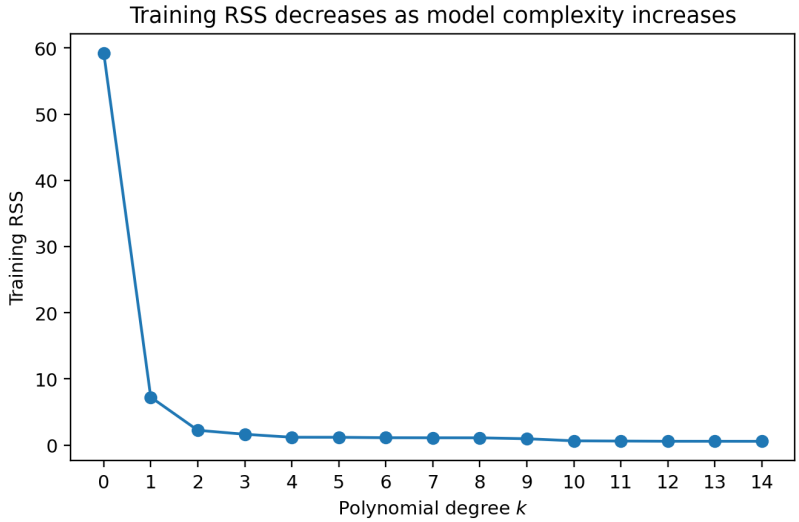


Figure 3: Training residual sum of squares as a function of polynomial degree. Because the degree-$(p+1)$ model contains the degree-p model as a special case, the training RSS cannot increase as the polynomial degree increases.

Thus, increasing the polynomial degree can only improve, or leave unchanged, the fit to the training data.

However,

smaller training error does not necessarily imply better prediction on new data.

A sufficiently flexible model can begin to fit sample-specific noise rather than the underlying conditional mean $\mu(x)$. Therefore, minimizing training RSS alone does not determine the appropriate polynomial degree. This motivates the need for model selection and regularization.

3 Bias and Variance of a Fitted Model

In the previous section, we saw that increasing the polynomial degree p enlarges the model class and therefore cannot increase the training residual sum of squares:

$$RSS_{p+1} \leq RSS_p.$$

However, this statement concerns only the particular training dataset that we observed.

A predictive model should also be stable with respect to changes in the training sample. To study this, suppose that we repeatedly collect new training datasets

$$D_1, D_2, \ldots$$

from the same underlying population and fit the same degree-p polynomial model to each dataset. Let

$$\hat{\mu}_{p,D}(x)$$

denote the fitted conditional mean at predictor value x obtained from training dataset D. The true conditional mean is

$$\mu(x) = \mathbb{E}[Y \mid X = x].$$

Because the observed training data are random, the fitted function

$$\hat{\mu}_{p,D}(x)$$

is also random: a different training dataset generally produces a different fitted curve.

The average fitted value across repeated training datasets is

$$\mathbb{E}_D\left[\hat{\mu}_{p,D}(x)\right].$$

The bias of the degree-p model at x is

$$\boxed{\text{Bias}_p(x) = \mathbb{E}_D\left[\hat{\mu}_{p,D}(x)\right] - \mu(x).}$$

Bias therefore measures the systematic difference between the average fitted model and the true conditional mean.

The variance of the fitted model at x is

$$\boxed{\text{Var}_D\left[\hat{\mu}_{p,D}(x)\right] = \mathbb{E}_D\left[\left(\hat{\mu}_{p,D}(x) - \mathbb{E}_D[\hat{\mu}_{p,D}(x)]\right)^2\right].}$$

Variance measures how much the fitted prediction changes when the training dataset changes.

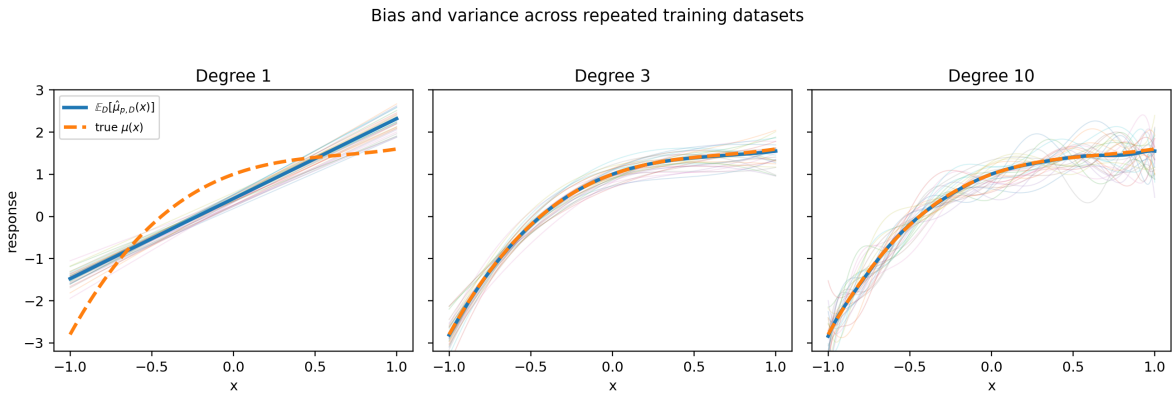


Figure 4: Bias and variance across repeated training datasets. Each thin curve is obtained by fitting the same polynomial model to a different training sample. The degree-1 model is stable across datasets but systematically misses the nonlinear conditional mean, giving high bias and low variance. The degree-3 model captures the conditional mean with relatively small bias and variance. The degree-10 model is more sensitive to the particular training sample, producing larger variance.

This illustrates the two competing effects of model complexity:

low complexity	$\Longrightarrow$	larger bias, smaller variance,
high complexity	$\Longrightarrow$	smaller bias, larger variance.

A simple model may be unable to represent the true conditional mean, while a highly flexible model may change substantially from one training sample to another.

3.1 Bias–Variance Decomposition

We now connect bias and variance to prediction error. Suppose a new response at predictor value x is generated according to

$$Y^* = \mu(x) + \varepsilon^*,$$

with

$$\mathbb{E}[\varepsilon^* \mid X = x] = 0, \qquad \mathrm{Var}(\varepsilon^* \mid X = x) = \sigma^2.$$

Here, Y^* is independent of the training dataset D. We want to understand the expected squared prediction error

$$\mathbb{E}_{D,Y^*}\left[(Y^* - \hat{\mu}_{p,D}(x))^2\right].$$

Substitute

$$Y^* = \mu(x) + \varepsilon^* :$$

$$Y^* - \hat{\mu}_{p,D}(x) = \mu(x) - \hat{\mu}_{p,D}(x) + \varepsilon^*.$$

Squaring and taking expectations,

$$\mathbb{E}_{D,Y^*}\left[(Y^* - \hat{\mu}_{p,D}(x))^2\right] = \mathbb{E}_D\left[(\mu(x) - \hat{\mu}_{p,D}(x))^2\right] + \sigma^2,$$

because the cross term vanishes:

$$\mathbb{E}[\varepsilon^*] = 0.$$

Now add and subtract

$$\mathbb{E}_D[\hat{\mu}_{p,D}(x)]$$

inside the first term:

$$\mu(x) - \hat{\mu}_{p,D}(x) = \mu(x) - \mathbb{E}_D[\hat{\mu}_{p,D}(x)] + \mathbb{E}_D[\hat{\mu}_{p,D}(x)] - \hat{\mu}_{p,D}(x).$$

Taking the expectation of the square gives

$$\mathbb{E}_{D,Y^*}\left[(Y^* - \hat{\mu}_{p,D}(x))^2\right] = \sigma^2 + \mathrm{Bias}_p(x)^2 + \mathrm{Var}_D\left[\hat{\mu}_{p,D}(x)\right].$$

The three terms have distinct meanings:

σ^2	irreducible variability in the response,
$\mathrm{Bias}_p(x)^2$	systematic error caused by model restriction,
$\mathrm{Var}_D[\hat{\mu}_{p,D}(x)]$	sensitivity of the fitted model to the training sample.

Increasing p therefore does not necessarily improve prediction on new data. Although training RSS decreases, the variance of the fitted model may increase.

This creates the central model-selection problem:

How can we retain a flexible model while reducing its sensitivity to the training data?

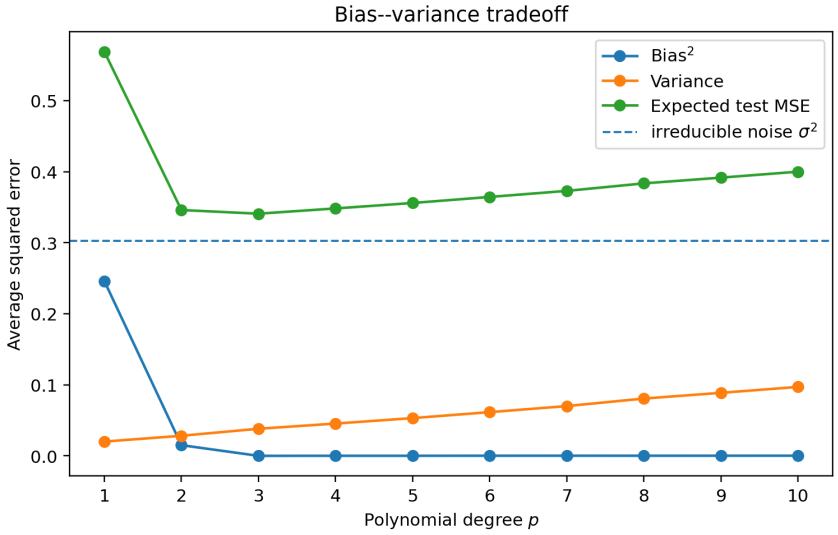


Figure 5: Bias–variance tradeoff as polynomial degree increases. Low-degree models have larger squared bias, while higher-degree models have larger variance. The expected prediction error combines squared bias, variance, and the irreducible noise σ^2.

4 Motivation for Regularization: Ridge regression

One option is to reduce the polynomial degree p. However, this also removes basis functions and therefore restricts the class of nonlinear relationships that the model can represent.

An alternative is to retain a rich set of basis functions,

$$1, x, x^2, \ldots, x^p,$$

but constrain the fitted coefficients so that the model does not use this flexibility excessively. For the degree-p model,

$$\mu_p(x) = \beta_0 + \beta_1 x + \cdots + \beta_p x^p,$$

ordinary least squares chooses

$$\hat{\beta}_{\text{OLS}} = \arg\min_{\beta} \|\mathbf{y} - X\beta\|_2^2.$$

When the basis is highly flexible, two related problems can occur:

large sampling variance	the fitted coefficients can change strongly across datasets,
poor conditioning	$X^T X$ can become nearly singular.

Regularization addresses both problems by discouraging excessively large coefficient values while retaining the full basis expansion. This leads to ridge regression:

$$\hat{\beta}_\lambda = \arg\min_{\beta} \left\{ \|\mathbf{y} - X\beta\|_2^2 + \lambda \|\beta\|_2^2 \right\}.$$

The first term favors agreement with the observed data, while the second term favors smaller coefficients. The regularization parameter $\lambda \geq 0$ controls this tradeoff between the two competing terms:

$$\underbrace{\|\mathbf{y} - X\beta\|_2^2}_{\text{fit to the training data}} + \lambda \underbrace{\|\beta\|_2^2}_{\text{penalty on coefficient magnitude}}.$$

Since

$$\|\beta\|_2^2 = \beta^T \beta = \sum_{j=0}^{p} \beta_j^2,$$

the ridge penalty discourages solutions with large coefficient norm.

If $\lambda = 0$, then the penalty disappears and

$$\hat{\beta}_\lambda = \hat{\beta}_{\text{OLS}}.$$

As λ increases, larger coefficient values become increasingly expensive, so the fitted coefficients are progressively shrunk toward zero. Thus,

λ controls the tradeoff between fitting the training data and shrinking the coefficients.

This shrinkage introduces bias, but can reduce the variance of the fitted model. Ridge therefore implements the bias–variance tradeoff discussed in the previous section:

$$\boxed{\text{increase bias slightly} \quad \Longrightarrow \quad \text{reduce variance and improve stability.}}$$

4.1 Equivalent Constrained Formulation

Ridge regression can also be written as a constrained optimization problem:

$$\boxed{\min_\beta \|\mathbf{y} - X\beta\|_2^2 \qquad \text{subject to} \qquad \|\beta\|_2^2 \le \delta^2.}$$

Here, $\delta > 0$ specifies the largest allowed coefficient norm. The constraint

$$\|\beta\|_2^2 \le \delta^2$$

restricts the coefficient vector to lie inside an L_2 ball. The Lagrangian for this constrained problem is

$$\mathcal{L}(\beta, \lambda) = \|\mathbf{y} - X\beta\|_2^2 + \lambda \left(\|\beta\|_2^2 - \delta^2 \right),$$

with

$$\lambda \ge 0.$$

For fixed λ, the term

$$-\lambda\delta^2$$

does not depend on β. Therefore, minimizing the Lagrangian with respect to β is equivalent to minimizing

$$\|\mathbf{y} - X\beta\|_2^2 + \lambda\|\beta\|_2^2.$$

Thus, for an appropriate correspondence between λ and δ, the penalized and constrained formulations produce the same ridge solution.

The two parameters describe the same idea in different ways:

$\lambda \uparrow$	$\Longleftrightarrow$	$\delta \downarrow$
stronger penalty		smaller feasible region.

4.2 Geometric Interpretation

Consider the two-parameter case

$$\beta = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}.$$

The ridge constraint becomes

$$\boxed{\beta_1^2 + \beta_2^2 \le \delta^2,}$$

which defines a disk centered at the origin in (β_1, β_2) space. The residual sum of squares

$$\|\mathbf{y} - X\beta\|_2^2$$

is a quadratic function of β. Its constant-value contours are ellipses centered at the ordinary least-squares solution

$$\hat{\beta}_{\text{OLS}}.$$

Without the constraint, the minimum occurs at the center of these ellipses. With the ridge constraint, only coefficient vectors inside

$$\|\beta\|_2^2 \leq \delta^2$$

are allowed. The ridge solution is therefore the point in the feasible region with the smallest possible residual sum of squares. Geometrically, it is the point where the smallest RSS contour first touches the constraint boundary.

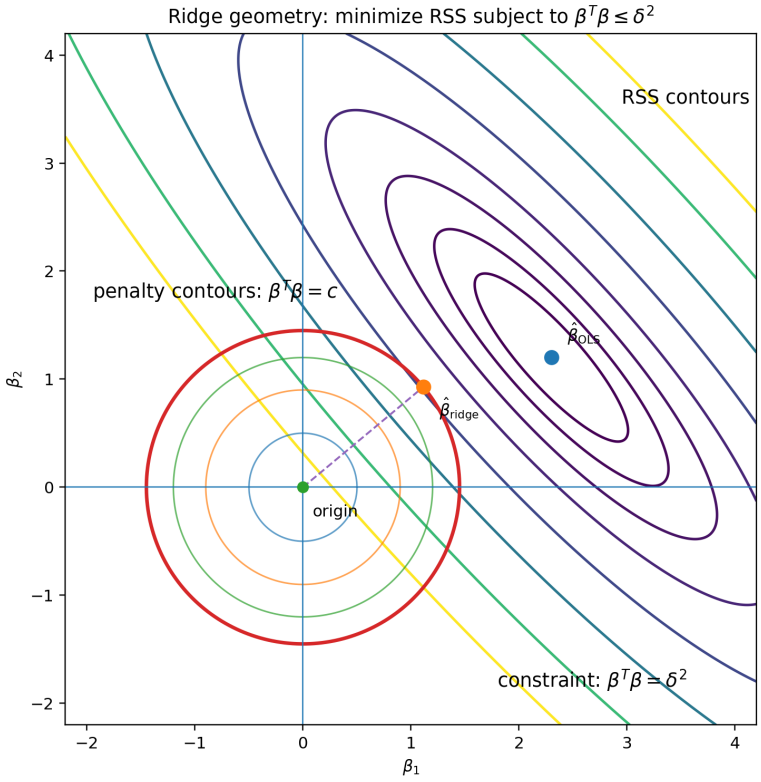


Figure 6: Geometric interpretation of ridge regression. The ellipses are contours of constant residual sum of squares, centered at the unconstrained OLS solution. The concentric circles are contours of $\beta^T\beta$. Ridge restricts the coefficients to the L_2 ball $\beta^T\beta \leq \delta^2$ and selects the point where the smallest attainable RSS contour touches the feasible region.

As δ decreases, the feasible region becomes smaller and the solution is forced closer to the origin. Equivalently, increasing λ produces stronger coefficient shrinkage.

4.3 Ridge Regression Solution

Define the ridge objective

$$J_\lambda(\beta) = \|\mathbf{y} - X\beta\|_2^2 + \lambda\|\beta\|_2^2.$$

Using

$$\|\mathbf{y} - X\beta\|_2^2 = (\mathbf{y} - X\beta)^T(\mathbf{y} - X\beta),$$

and

$$\|\beta\|_2^2 = \beta^T\beta,$$

we obtain

$$J_\lambda(\beta) = (\mathbf{y} - X\beta)^T(\mathbf{y} - X\beta) + \lambda\beta^T\beta.$$

Expanding,

$$J_\lambda(\beta) = \mathbf{y}^T\mathbf{y} - 2\beta^T X^T\mathbf{y} + \beta^T X^T X\beta + \lambda\beta^T\beta.$$

Differentiating with respect to

$$\beta \in \mathbb{R}^{(p+1)\times 1}$$

gives

$$\nabla_\beta J_\lambda = -2X^T\mathbf{y} + 2X^T X\beta + 2\lambda\beta.$$

At the minimum,

$$\nabla_\beta J_\lambda = 0.$$

Therefore,

$$-2X^T\mathbf{y} + 2X^T X\hat{\beta}_\lambda + 2\lambda\hat{\beta}_\lambda = 0,$$

which gives the ridge normal equations

$$(X^T X + \lambda I_{p+1})\hat{\beta}_\lambda = X^T\mathbf{y}.$$

Hence,

$$\hat{\beta}_\lambda = (X^T X + \lambda I_{p+1})^{-1}X^T\mathbf{y}.$$

The dimensions are

$$X^T X \in \mathbb{R}^{(p+1)\times(p+1)},$$

$$\lambda I_{p+1} \in \mathbb{R}^{(p+1)\times(p+1)},$$

and

$$X^T\mathbf{y} \in \mathbb{R}^{(p+1)\times 1},$$

so

$$\hat{\beta}_\lambda \in \mathbb{R}^{(p+1)\times 1}.$$

The important change relative to ordinary least squares is therefore

$$\boxed{X^T X \quad \longrightarrow \quad X^T X + \lambda I.}$$

This modification has two consequences:

statistical:	the coefficients are shrunk, reducing variance,
numerical:	the matrix inversion becomes better conditioned.

NOTE: For simplicity we penalize all coefficients here; in practice, the intercept is typically left unpenalized after centering the predictors and response. The next question is therefore why adding λI improves the conditioning of the estimation problem.

4.4 Why Does Ridge Improve Conditioning?

Let

$$X^T X = Q \Lambda Q^T,$$

where

$$\Lambda = \mathrm{diag}(\gamma_1, \ldots, \gamma_{p+1})$$

contains the eigenvalues of $X^T X$. Since Q is orthogonal,

$$Q Q^T = I,$$

and therefore

$$\lambda I = Q(\lambda I) Q^T.$$

Hence,

$$\begin{aligned} X^T X + \lambda I &= Q \Lambda Q^T + Q(\lambda I) Q^T \\ &= Q(\Lambda + \lambda I) Q^T. \end{aligned}$$

Thus, adding λI leaves the eigenvectors unchanged and shifts each eigenvalue from γ_j to

$$\gamma_j + \lambda.$$

If one eigenvalue of $X^T X$ is very small,

$$\gamma_{\min} \approx 0,$$

then the matrix is poorly conditioned. Ridge shifts every eigenvalue away from zero:

$$\gamma_{\min} \quad \longrightarrow \quad \gamma_{\min} + \lambda.$$

Hence,

$$\boxed{\kappa(X^T X + \lambda I) = \frac{\gamma_{\max} + \lambda}{\gamma_{\min} + \lambda}.}$$

As λ increases, this condition number decreases toward 1.

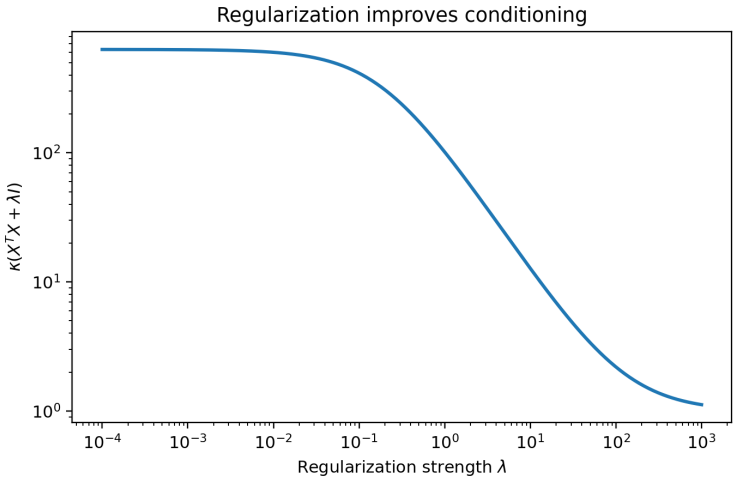


Figure 7: Ridge regularization improves conditioning by shifting the eigenvalues of $X^T X$ away from zero.

4.5 Shrinkage of the Coefficients

Because

$$\hat{\beta}_\lambda = (X^T X + \lambda I)^{-1} X^T \mathbf{y},$$

every value of λ produces a different fitted coefficient vector. As

$$\lambda \to \infty,$$

the penalty dominates the objective and $\hat{\beta}_\lambda \to 0$. The trajectories $\hat{\beta}_j(\lambda)$ are called *regularization paths*.

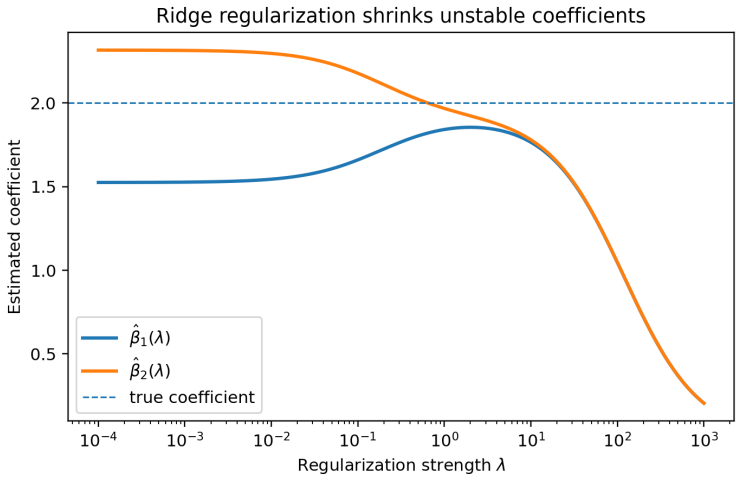


Figure 8: Regularization path for two strongly correlated predictors. Increasing λ shrinks the fitted coefficients toward zero and reduces their instability.

We therefore have two competing limits:

$$\lambda \approx 0 \implies \text{weak regularization, flexible fit, larger variance,}$$
$$\lambda \gg 1 \implies \text{strong regularization, strong shrinkage, larger bias.}$$

This introduces the next problem:

$$\boxed{\text{How should we choose } \lambda?}$$

5 Model selection: How to tune the parameter λ:

5.1 Training and Validation Error

Suppose the available data are divided into two disjoint sets,

$$D_{\text{train}} \qquad \text{and} \qquad D_{\text{val}}.$$

The training set is used to estimate the regression coefficients for a chosen value of λ. For each candidate λ, fit $\hat{\beta}_\lambda$ using only

$$D_{\text{train}}.$$

The resulting fitted conditional mean is denoted $\hat{\mu}_\lambda(x)$. If the training set contains n_{tr} observations, define

$$\text{MSE}_{\text{train}}(\lambda) = \frac{1}{n_{\text{tr}}} \sum_{i \in D_{\text{train}}} \left(y_i - \hat{\mu}_\lambda(x_i)\right)^2.$$

The validation set is not used to estimate $\hat{\beta}_\lambda$. It is used only to evaluate how well the fitted model predicts unseen observations:

$$\mathrm{MSE}_{\mathrm{val}}(\lambda) = \frac{1}{n_{\mathrm{val}}} \sum_{i \in D_{\mathrm{val}}} \left(y_i - \hat{\mu}_\lambda(x_i)\right)^2 .$$

We then choose

$$\boxed{\lambda^* = \arg\min_\lambda \mathrm{MSE}_{\mathrm{val}}(\lambda).}$$

The roles of the two errors are therefore different:

$\mathrm{MSE}_{\mathrm{train}}$	measures fit to the data used to estimate the coefficients,
$\mathrm{MSE}_{\mathrm{val}}$	measures predictive performance on held-out data.

Validation therefore provides a data-driven way to choose the regularization strength.

5.2 Why a Single Train–Validation Split Can Be Unstable

A single train–validation split introduces another source of randomness: the particular observations assigned to each subset. Suppose the same dataset is split in two different ways,

$$D = D^{(1)}_{\mathrm{train}} \cup D^{(1)}_{\mathrm{val}}$$

and

$$D = D^{(2)}_{\mathrm{train}} \cup D^{(2)}_{\mathrm{val}}.$$

The fitted models are generally different because the training sets are different. Consequently,

$$\mathrm{MSE}^{(1)}_{\mathrm{val}}(\lambda)$$

and

$$\mathrm{MSE}^{(2)}_{\mathrm{val}}(\lambda)$$

need not have their minima at the same value of λ. Thus,

$$\boxed{\lambda^* \text{ can depend strongly on the particular train–validation split.}}$$

This effect is especially important when the dataset is small or noisy. A more stable estimate of predictive performance should use the available data more systematically. This motivates K-fold cross-validation.

5.3 K-Fold Cross-Validation

Divide the available model-development data into K disjoint subsets,

$$D_1, \ldots, D_K,$$

called *folds*. For fold k, define

$$D_{-k} = \bigcup_{j \neq k} D_j.$$

The model is fitted using D_{-k}, while

$$D_k$$

is held out for validation. For a candidate value of λ, let

$$\hat{\mu}^{(-k)}_\lambda(x)$$

denote the fitted conditional mean obtained using all folds except D_k. The validation error on fold k is

$$E_k(\lambda) = \frac{1}{|D_k|} \sum_{i \in D_k} \left(y_i - \hat{\mu}^{(-k)}_\lambda(x_i)\right)^2 .$$

Repeating this procedure for

$$k = 1, \ldots, K$$

produces K validation errors. The K-fold cross-validation error is their average:

$$CV_K(\lambda) = \frac{1}{K} \sum_{k=1}^{K} E_k(\lambda).$$

We then choose

$$\lambda^* = \arg\min_{\lambda} CV_K(\lambda).$$

Each observation is therefore

used for validation exactly once

and

used for training in the remaining $K - 1$ folds.

Compared with a single random split, cross-validation reduces the dependence of model selection on one particular choice of validation data.

5.4 Cross-Validation for Model Complexity

Cross-validation is not specific to ridge regression. More generally, it can be used to choose any hyperparameter that controls model complexity. For example, consider polynomial regression models of degree

$$p = 1, 2, \ldots, p_{\max}.$$

For each degree p, fit the model within each training fold and compute

$$CV_K(p).$$

The selected degree is

$$\boxed{p^* = \arg\min_{p} CV_K(p).}$$

Thus, the same principle applies whether model complexity is controlled by

p	polynomial degree,
λ	regularization strength,
number of basis functions	size of the feature representation.

In each case, the hyperparameter is chosen using predictive performance on data that were not used to estimate the model coefficients. The resulting general principle is

fit parameters on training data $\longrightarrow$ select hyperparameters (λ) using the validation data.

5.5 Training, Validation, and Test Data

The roles of the three datasets must be kept distinct.

Training data	$\longrightarrow$	estimate model parameters such as β,
Validation data	$\longrightarrow$	select hyperparameters such as λ or p,
Test data	$\longrightarrow$	evaluate the final selected model.

The test set must not be used to choose

$$\beta, \qquad \lambda, \qquad p,$$

or any other aspect of modeling selection. Otherwise, the test set becomes part of the model-selection procedure and no longer provides an independent estimate of predictive performance. After selecting the hyperparameters, the model may be refitted using all available non-test data before it is evaluated once on the test set.

Summary

The model-building problem can now be written as a sequence:

choose basis functions → fit parameters → control complexity → select hyperparameters → evaluate on unseen data.

More explicitly,

Basis expansion	allows nonlinear dependence while remaining linear in β,
Regularization	stabilizes flexible models and shrinks coefficients,
Validation	selects model complexity,
K-fold CV	reduces sensitivity to one train–validation split,
Test data	estimates final predictive performance,